INCH-POUND

MIL-DTL-85097/5B(OS)
31 January 1996
SUPERSEDING
MIL-A-85097/5A(AS)
17 October 1985

ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTORS, MK 82 MODS 0 AND 1 (MAN/SEAT SEPARATOR, LEFT) AND MK 90 MODS 0 AND 1 (MAN/SEAT SEPARATOR, RIGHT) (LOADED ASSEMBLIES)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the motors described herein shall consist of this specification and the issue of MIL-A-85097 listed in the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation.

1. SCOPE

1.1 Scope. This associated detail specification covers requirements unique to the Aircrew Escape Propulsion Systems (AEPS) Rocket Motors, MK 82 Mods 0 and 1 (Man/Seat Separator, Left) and MK 90 Mods 0 and 1 (Man/Seat Separator, Right) (Loaded Assemblies), hereinafter referred to as the "rocket motors" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standard, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-A-85097 Aircrew Escape Propulsion Systems (AEPS), Procurement Specification For

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

AS 3550 Propellant, N-43 (U)

NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

511-174TD-004 Propellant, NOSIH-BC-11, NSWC/IH-EC-02, NSWC/IH-EC-04, And NSWC/IH-EC-05

STANDARD

FEDERAL

FED-STD-595 Colors Used In Government

(Unless otherwise indicated, copies of Department of Defense Specifications and Standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

944AS100 Rocket Motor, MK 82 Mod 0 (Man/Seat Separator, (Left))

944AS212 Rocket Motor, MK 82 Mod 1 (Man/Seat Separator, (Left))

970AS100 Rocket Motor, MK 90 Mod 0 (Man/Seat Separator, (Right))

970AS201 Rocket Motor, MK 90 Mod 1 (Man/Seat Separator, (Right))

(Application for copies of Navy specifications and drawings should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Order of precedence. The requirements in this specification, unless in direct conflict with the general specification, shall be considered additional, not substitute. If there is a direct conflict with the general specification, the associated detail specification shall take precedence.

3. REQUIREMENTS

3.1 General. The MK 82 and MK 90 Mod 0 Rocket Motors shall conform to Drawings 944AS100 and 970AS100 respectively. The MK 82 and MK 90 Mod 1 Rocket Motors shall conform to Drawings 944AS212 and 970AS201 respectively.

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3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the Mod 0 propellant shall meet the aging requirements AS 3550 and the Mod 1 propellant shall meet the aging requirements specified in 511-174TD-004 (see 6.3).

3.3 Preproduction/production.

3.3.1 Propellant. The Mod 0 propellant shall conform to the requirements of AS 3550 and the Mod 1 propellant shall conform to the requirements of 511-174TD-004 when tested in accordance with MIL-A-85097. The contractor shall submit all test results in accordance with AS3550, 511-174TD-004, and MIL-A-85097.

3.3.2 Ballistic performance. The ballistic performance shall conform to table I.

TABLE I. Ballistic performance requirements.

Parameter	-40° ± 5°F		70° ± 5°F		160° ± 5°F	
	Min	Max	Min	Max	Min	Max
Ignition delay (ms)	—	10	—	10	—	10
Action time (ms)	—	263[1]	—	235	—	212
Rocket motor resultant thrust (max)	439[1]	725	510	840	570	950[1]
Rocket motor impulse (lb-s)	97[1]	—	97[1]	—	97[1]	—
Time to 375-pound thrust (ms)	—	26[1]	—	26[1]	—	26[1]
Burn time (ms)	163	—	140	—	120[1]	—

[1] System requirements.

3.3.3 Finish. Color shall be dark gull gray, No. 36231 in accordance with FED-STD-595.

4. VERIFICATION

4.1 Preproduction/production.

4.1.1 Conditioning temperatures. The conditioning temperatures shall be as follows:

a. Cold: -40 ± 5°F
b. Hot: 160 ± 5°F

4.1.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 5 to 72 hours
b. Temperature cycling: 5 to 7 hours

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4.1.3 Out of condition limits. Once the rocket motor is removed from the conditioning chamber for testing, or if the rocket motor temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket motor shall be reconditioned. A reconditioning time of 2 minutes is required for every minute the rocket motor spends outside the specified temperature envelope. Reconditioning time shall not exceed 3 hours.

5. PACKAGING

5.1 Packing. Rocket motors shall be packaged per MIL-A-85097 and the applicable contract (see 6.2).

6. NOTES

(This section contains information of a general nature or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Rocket Motors Mk 82 Mods 0 and 1 are intended for use as a means of ensuring positive separation of an aircrewman from his ejection seat for the S-3, A-4, and A-7 aircraft escape systems. The Rocket Motors Mk 90 Mods 0 and 1 are used only in the S-3 aircraft.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

6.2 Configurations. The Rocket Motors Mk 82 Mod 0 and Mk 90 Mod 0, which use N-43 propellant, shall not be used for new production. The Mod 1 configuration shall be used for all new production.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy - OS
(Project 1377-0036)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-DTL-85097/5B(OS)	960131

3. DOCUMENT TITLE AIRCREW ESCAPE PROPULSTION SYSTEMS (AEPS); ROCKET MOTORS, MK82 MODS) AND 1 (MAN/SEAT SEPARATOR, LEFT) AND MK 90 MODS 0 AND 1 (MAN/SEAT SEPARATOR, RIGHT) LOADED ASSEMB.

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8420)	(1) Commercial 301-743-4358/4510	(2) AUTOVON 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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